

Reference Guide

Introduction

Target Devices

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1. Overview

1.1 Radio Driver Component Diagram

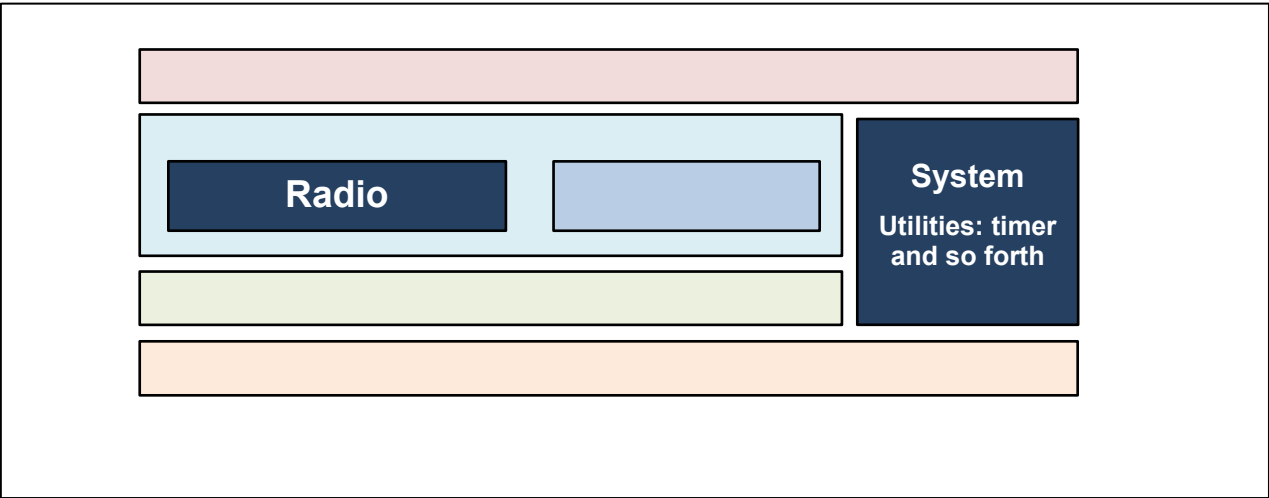


Figure 1. Radio Driver and Related Components Overview

1.2 Directories (informative)

Table 1. Directories

Directories	Description

1.3 Resource Usage

- *RL78/G23, RL78/G22, RL78/L23, RL78/G14 LoRa®-based Wireless Software Package*
- *RA2E1, RA2L1, RA0E1, RA0E2 LoRa®-based Wireless Software Package*

1.4 Related Documentation

	No.	Title	Author	Language

2. Radio Interface (LoRa / (G)FSK)

Type / Functions	Description

Type / Functions	Description

2.1 Defines

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2.1.2 Misc

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2.2.1 RADIO_WAKEUP_TIME

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2.3 Enumeration

2.3.1 RadioModems_t

2.3.2 RadioState_t

2.3.3 RadioTcxoCtrlVoltage_t

2.3.4 PIB_t

2.3.5 RadioResult_t

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2.3.7 RadioConfigRegion_t

2.4 Type definition

2.5 Structure

2.5.1 RadioEvents_t

2.5.2 RadioTxFailStatus_t

[illegible]

2.6.1 Init

RadioResult_t (RADIO_FAR *Init)(RadioEvents_t *event)			

```
RadioResult_t ( RADIO_FAR *SetRxConfig ) ( RadioModems_t modem, uint32_t bandwidth,
uint32_t datarate, uint8_t coderate, uint32_t bandwidthAfc, uint16_t preambleLen,
uint16_t symbTimeout, bool fixLen, uint8_t payloadLen, bool crcOn, bool FreqHopOn,
uint8_t HopPeriod, bool iqlInverted, bool rxContinuous )
```

[illegible]

2.6.3 SetTxConfig

RadioResult_t (RADIO_FAR* SetTxConfig)(RadioModems_t modem, int8_t power, uint32_t fdev, uint32_t bandwidth, uint32_t datarate, uint8_t coderate, uint16_t preambleLen, bool fixLen, bool crcOn, bool FreqHopOn, uint8_t HopPeriod, bool iqlInverted, uint32_t timeout)		

2.6.6 SetMaxPayloadLength

void (RADIO_FAR *SetMaxPayloadLength)(RadioModems_t modem, uint8_t max)	

2.6.7 SetPublicNetwork

void (RADIO_FAR *SetPublicNetwork)(bool enable)	

2.6.8 Rx

RadioResult_t (RADIO_FAR *Rx)(uint32_t timeout)	

2.6.9 Send

RadioResult_t (RADIO_FAR *Send)(uint8_t RADIO_FAR *buffer, uint8_t size)	

2.6.10 Sleep / SleepWarm

void (RADIO_FAR *Sleep / *SleepWarm)(void)	

2.6.11 Standby

void (RADIO_FAR *Standby)(void)	

2.6.12 SetTxContinuousWave

RadioResult_t (RADIO_FAR *SetTxContinuousWave)(uint32_t freq, int8_t power, uint16_t time)	

2.6.13 SetTxInfinitePreamble

RadioResult_t (RADIO_FAR *SetTxInfinitePreamble)(uint32_t freq, int8_t power, uint16_t time)	

2.6.14 Rssi

int16_t (RADIO_FAR *Rssi)(RadioModems_t modem)	

2.6.15 TimeOnAir

uint32_t (RADIO_FAR *TimeOnAir)(RadioModems_t modem, uint8_t pktLen)	

2.6.16 CheckRfFrequency

bool (*CheckRfFrequency)(uint32_t frequency)	

2.6.17 GetStatus

RadioState_t (RADIO_FAR *GetStatus)(void)	

2.6.18 IsChannelFree

bool (RADIO_FAR *IsChannelFree)(RadioModems_t modem, uint32_t freq, int16_t rssiThresh,

2.6.22 GetPib

bool (RADIO_FAR *GetPib)(PIB_t id, uint8_t RADIO_FAR * pOutVal)		

2.6.23 SetPib

bool (RADIO_FAR *SetPib)(PIB_t id, uint8_t RADIO_FAR * pInVal)		

2.6.24 WakeUp

void (RADIO_FAR *WakeUp)(void)		

2.6.25 SleepCold

void (RADIO_FAR *SleepCold)(void)		

2.6.26 CalibrateImage

void (RADIO_FAR *CalibrateImage)(uint32_t freq)		

[illegible]

void (*Write)(uint16_t addr, uint8_t data)	

uint8_t (*Read)(uint16_t addr)	

[illegible]

uint32_t (*GetWakeupTime)(void)	

[illegible][illegible]

void (RADIO_FAR *RxDone)(uint8_t *payload, uint16_t size, int16_t rssi, int8_t snr)	

2.7.4 RxTimeout

void (RADIO_FAR *RxTimeout)(void)

2.7.5 RxError

void (RADIO_FAR *RxTimeout)(void)

2.8 Radio Driver API Constraint

2.8.1 Modem parameter

API	Description

[illegible]

3. Timer

3.1 Type Definition

Table 4. Type Definition

Type	Description

3.1.1 TimerEvent_t

3.1.2 TimerTime_t

Table 5. TimerTime_t

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3.2 Timer APIs

Function	Description

3.2.1 TimerInit

void TimerInit(TimerEvent_t *obj, void (*callback)(void))	

3.2.2 TimerSetValue

void TimerSetValue(TimerEvent_t *obj, uint32_t value)	

3.2.3 TimerStart

void TimerStart(TimerEvent_t *obj)	

3.2.4 TimerStop

void TimerStop(TimerEvent_t *obj)	

3.2.5 TimerReset

void TimerReset(TimerEvent_t *obj)	

3.2.6 TimerGetCurrentTime

TimerTime_t TimerGetCurrentTime(void)	

3.2.7 TimerGetElapsedTime

TimerTime_t TimerGetElapsedTime(TimerTime_t savedTime)	

General Precautions in the Handling of Microprocessing Unit and Microcontroller Unit Products

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